Honours Thesis - Research Block 2

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I. INTRO

DO NOT COPY ANYTHING FROM HERE INTO THE THESIS!!! MANY SENTENCES ARE DIRECT QUOTES!!!

II. MICROSCOPIC THEORY OF SQUEEZED LIGHT IN QUANTUM DOT SYSTEMS

'In practice, amplitude squeezing is maintained only if classical pump relative-intensity noise (RIN) is suppressed below shot noise.' This is important experimental note

'Simulations that average over a distribution of injected frequencies (to model finite laser linewidth) show that squeezing remains essentially unaffected so long as the injected linewidth is roughly three orders of magnitude smaller than the dephasing rate.'

'stronger squeezing demands narrower effective linewidths (smaller γ_c), reflecting the familiar tradeoff between output power and preservation of quantum correlations.'

'From an engineering perspective, three levers govern performance, cavity decay, seed strength, and dephasing.'

IMPORTANT EQUATIONS:

$$\Delta X^2 = \frac{1}{4}(1 + 2\delta\langle a^\dagger a\rangle + 2Re(\delta\langle aa\rangle))$$

$$\Delta Y^2 = \frac{1}{4}(1 + 2\delta\langle a^\dagger a\rangle - 2Re(\delta\langle aa\rangle))$$

$$\delta\langle a^\dagger a\rangle = \langle a^\dagger a\rangle - \langle a^\dagger\rangle\langle a\rangle$$

$$\delta\langle aa\rangle = \langle aa\rangle - \langle a\rangle\langle a\rangle$$

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger aa\rangle}{\langle a^\dagger a\rangle\langle a^\dagger a\rangle}$$